

YOU HAVE YOUR ZONES. THIS IS WHAT TO DO WITH THEM.

The sessions that grow each zone

Swimming, cycling and running



Having the numbers is half of it. The other half is knowing which session builds each zone — how to put together a Zone 2 that's more than easy miles, the marathon-pace session that justifies Zone X, and what a real threshold set looks like.

How to read the numbers

The model

Polarised 80/20: roughly 80% of your volume stays easy and 20% is genuinely hard. Five zones would cover that. The other two exist for a different reason.

X and Y aren't new zones

They are **pieces carved out of Zone 2 and Zone 4**. We gave them names because we want to avoid both — and you can't avoid a zone that doesn't have a name. The interesting part is that we avoid them for exactly opposite reasons.

Zone X — the fast part of Zone 2

Zone X is the ceiling of your Zone 2. It costs considerably more in fatigue and hands back almost the same aerobic stimulus you'd have got going slower: you pay more and take home the same.

It's the most common mistake in endurance training, and it's almost never a decision. You head out for an easy run, you feel good, the pace settles a few seconds quicker on its own, and you end the week carrying the fatigue of a quality session you never did — with nothing left for the one you had actually planned.

Marathon and Ironman are the exception: there Zone X is your race pace, and it has to be trained on purpose.

Zone Y — the floor of Zone 4

Zone Y is avoided too, but the intuition runs the other way: it isn't too much, it's **not enough**.

Your threshold isn't a fixed line. It moves with fitness, rest, heat and the day. So training at exactly 100% of your number guarantees nothing: if your real threshold sat a little higher that day, the session came up short and the adaptations never arrived. You paid the cost of a threshold session without buying the benefit.

To be sure of them you have to train slightly above: **from 102%**, which is precisely the floor of Zone 4. Zone Y is that band of doubt between 100 and 102 — trained only when it's your exact race pace.

Every percentage is of your threshold

Not of your maximum heart rate. And they don't agree between metrics: pace, heart rate and power behave differently at the extremes. That's why each sport has its own table.

Don't have your zones yet?

The calculator gives them to you free from your test result, in all three sports: [triaperformance.com/en/training-zones-calculator](https://www.triaperformance.com/en/training-zones-calculator)

Running

Primary metric: power > pace > heart rate

Heart rate lags behind the effort, so it's the backup rather than the main signal whenever a better metric is available. The exception is easy days, where HR works well as a ceiling.

ZONE	% HR (LTHR)	% PACE	WHAT IT'S FOR
1 · Recovery	72–81%	60–76%	Circulation, mobility, active days off
2 · Aerobic base	81–90%	76–87%	Where 80% of your volume lives
X · High aerobic	90–95%	87–93%	Marathon pace
3 · Tempo	95–100%	93–100%	Extensive threshold. Key session
Y · Low threshold	100–102%	100–102%	The band right around threshold
4 · High threshold	102–105%	102–115%	Intensive threshold. The other key session
5 · VO2max	105–120%	115–140%	Short reps, low volume, high cost

1	Recovery Circulation, mobility, active days off	% HR (LTHR)	% PACE
		72–81%	60–76%

The lowest intensity there is. It feels almost uncomfortably slow and takes deliberate holding back.

WHAT IT TRAINS

Moves blood through damaged muscle and speeds up the clearing of metabolic waste without introducing new stress that itself needs recovering from.

SESSIONS

1. **Recovery jog** — 20 to 40 very easy minutes, the day after a hard session.
2. **Warm-up and cool-down** — 10 to 15 minutes either side of any quality session.

WHAT NOT TO DO

Don't look at the watch hunting for a respectable pace. If you're worried the pace looks slow — or about what shows up on Strava — you'll push into Z2 or ZX and the day loses its purpose.

~20–25% of weekly time



Aerobic base

Where 80% of your volume lives

% HR (LTHR)

% PACE

81–90%

76–87%

Your natural easy-run pace. You have to be able to hold a full conversation without your breathing breaking up.

WHAT IT TRAINS

Mitochondrial development, more capillaries, better use of fat as fuel, resistance to fatigue. This is the zone that builds the engine.

SESSIONS

1. **Long run** — 90 to 120+ continuous minutes.
2. **Base run** — 45 to 60 minutes, midweek aerobic maintenance.
3. **Progression run** — 60 minutes: 15' in Z1 and the rest in Z2, finishing in the upper half of the zone without crossing out of it.

WHAT NOT TO DO

Don't push the hills or close with a sprint to the door. That spikes your heart rate and drops you into the moderate-intensity black hole.

~55–60% of weekly time



High aerobic

Marathon pace

% HR (LTHR)

% PACE

90–95%

87–93%

Marathon pace. It adds a lot more fatigue than Z2 without adding much more aerobic benefit. It gets prescribed on purpose or it doesn't get touched.

WHAT IT TRAINS

Specific resistance to fatigue at long-course race pace, and economy at that exact speed.

SESSIONS — MARATHON OR IRONMAN BLOCKS ONLY

1. **Long run with quality** — 45–60 minutes in Z2, then 40–50 minutes in Zone X, 10' easy to finish.
2. **Broken blocks** — 3 x 20 minutes in Zone X with 5' in Z1 between blocks, inside a long run.

WHAT NOT TO DO

Don't land here by accident on an easy day. If you're not building for a long-course race, avoid it.

0% outside a block; inside one, 5–10%, and it comes out of the hard budget



Tempo

Extensive threshold. Key session

% HR (LTHR)

% PACE

95–100%

93–100%

Comfortably hard. For most people it lands near half-marathon pace; the pace you could hold for a full hour sits at the top edge of the zone, not in the middle.

WHAT IT TRAINS

Raises your capacity to recycle lactate — which lifts the threshold — and improves resistance at submaximal speeds. One of the two key sessions of the week.

SESSIONS

1. **Continuous tempo** — 15' Z1 + 20 to 30 sustained minutes in Z3 + 10' Z1.
2. **Cruise intervals** — 4 x 8 minutes in Z3 with 2 minutes in Z1.
3. **Long broken tempo** — 3 x 15 minutes in Z3 with 3 minutes in Z1. Advanced: 45 minutes of work.

WHAT NOT TO DO

Turning the tempo into a race. Drift into Z4 and you lose the metabolic purpose and arrive at the next session tired.

~5–10%



Low threshold

The band right around threshold

% HR (LTHR)

% PACE

100–102%

100–102%

The razor-thin band right around threshold. Too demanding to hold as a tempo, and too gentle to produce the VO2max adaptations.

It's **15k** territory, and it's the **last 3 kilometres of a half marathon** — when there's nothing left to manage and the pace lifts on its own.

WHAT IT TRAINS

Tolerance for working exactly at threshold, and familiarity with that pace if it happens to be your race pace.

SESSIONS — ONLY IF YOU RACE AT THIS PACE

1. **Race rehearsal** — 3 x 10 minutes in Zone Y with 3' recovery.
2. **Half marathon from the finish backwards** — 30 continuous minutes in Z3 and then, without pausing, 10 minutes in Zone Y. It trains what actually decides a half: closing above threshold with 18 kilometres in your legs.
3. **Short blocks** — 5 x 5 minutes in Zone Y with 90 seconds easy.

WHAT NOT TO DO

Programming it as routine. If you're not preparing for a distance run at exactly this pace, Z3 and Z4 do the same job better spread out.

Residual outside race-specific blocks



High threshold

Intensive threshold. The other key session

% HR (LTHR)

% PACE

102–105%

102–115%

Intervals above threshold, longer than a VO2max rep and shorter than a tempo. Z3 accumulates volume at threshold; Z4 accumulates intensity just above it.

WHAT IT TRAINS

Raises the threshold from above rather than below: it improves your capacity to tolerate and recycle lactate at speeds you couldn't hold for half an hour.

SESSIONS

1. **Long threshold reps** — 5 x 6 minutes in Z4 with 2 minutes in Z1.
2. **Medium reps** — 8 x 3 minutes in Z4 with 90 seconds in Z1.
3. **Double blocks** — 2 x 12 minutes in the lower part of Z4 with 3 minutes in Z1.

WHAT NOT TO DO

Don't run it at 5k pace. Z4 by pace reaches 115%, but these sessions live in the lower half of that band: start too fast and the session becomes a half-hearted VO2max effort that does neither job.

~5–8%



VO2max

Short reps, low volume, high cost

% HR (LTHR)

% PACE

105–120%

115–140%

Short reps, low volume, high cost. 3k to 5k pace in the lower part of the zone; above that, pure neuromuscular work.

WHAT IT TRAINS

Raises VO2max, improves stroke volume, and sharpens running economy noticeably.

SESSIONS

1. **Classic VO2max set** — 6 x 800 metres with 2 to 3 minutes recovery.
2. **Timed intervals** — 5 x 3 minutes with 2 to 3 minutes recovery.
3. **Short reps** — 10 to 12 x 400 metres with 90 seconds to 2 minutes.

Neuromuscular work. 8 to 10 x 15 seconds hard uphill, walking back down, with 2 to 3 minutes between reps. **Prescribed by duration and maximal effort, not by zone** — the effort is over before your heart rate gets anywhere. They go at the end of an easy run, in very small doses.

WHAT NOT TO DO

Cutting the recoveries out of false heroism. Shorten the rest and you'll arrive at rep 5 so tired the pace falls to Z3. You have to be able to repeat the same pace on the last rep as on the first.

~2–5%, including the neuromuscular work

Cycling

Primary metric: power (FTP)

Power always leads. Heart rate trails it by several minutes and is useless for pacing short intervals. It is good for one thing power can't tell you: decoupling. If three hours in you're still at Z2 watts but your pulse has gone to Z3, your metabolic Zone 2 is over for the day.

ZONE	% FTP	% HR (LTHR)	WHAT IT'S FOR
1 · Recovery	50–70%	72–81%	Circulation, active days off
2 · Aerobic base	70–83%	81–90%	Where 80% of your volume lives
X · High aerobic	83–91%	90–95%	70.3 and Ironman pace
3 · Tempo	91–100%	95–100%	Extensive threshold. Key session
Y · Low threshold	100–102%	100–102%	The band right around FTP
4 · High threshold	102–110%	102–105%	Intensive threshold. The other key session
5 · VO2max	110–150%	105–120%	Short reps, low volume, high cost

1	Recovery	% FTP	% HR (LTHR)
	Circulation, active days off	50–70%	72–81%

The coffee-shop spin. Legs turning with no resistance.

WHAT IT TRAINS

Circulation to clear metabolic waste without recruiting fast fibre or adding central fatigue.

SESSIONS

1. **Recovery spin** — 30 to 45 minutes at high cadence (90+ rpm) with no resistance.
2. **Warm-up and cool-down** — 15 easy minutes either side of an interval block.

WHAT NOT TO DO

Pushing the rises. On the road, the temptation to stand up on a climb puts you in Z3 or Z4 within seconds. Outside, small ring and big sprocket; on the trainer, ERG mode.

~20–25%



Aerobic base

Where 80% of your volume lives

% FTP % HR (LTHR)

70–83% **81–90%**

Your endurance pace. The zone that builds the engine.

WHAT IT TRAINS

Mitochondrial biogenesis, capillary density and fat oxidation.

SESSIONS

1. **Long ride** — 2 to 4+ hours steady in the middle of Z2.
2. **Midweek ride** — 60 to 90 steady minutes.

WHAT NOT TO DO

Being reactive to the terrain. Cycling's most expensive mistake is a high variability index: coasting at zero watts downhill and hitting every rise in Z4. Pedal on the descents too, to hold muscular tension inside the zone.

~55–60%



High aerobic

70.3 and Ironman pace

% FTP % HR (LTHR)

83–91% **90–95%**

What cycling calls sweet spot. A lot of peripheral muscular stress per watt. It's the pace of a 70.3 and of an Ironman.

WHAT IT TRAINS

Specific resistance to fatigue and biomechanical efficiency in an aero position.

SESSIONS — 70.3 / IRONMAN SPECIFIC

1. **Long race-pace intervals** — 4 x 20 minutes in ZX with 5' easy. Advanced session: that's 80 minutes in zone.
2. **70.3 simulation** — 2 hours with 3 blocks of 30 minutes in ZX.

WHAT NOT TO DO

Turning the weekend group ride into this. Those rides tend to be an uninterrupted party in Zone X: they burn a lot and make you very little faster.

0% outside a 70.3/IM block



Tempo

Extensive threshold. Key session

% FTP % HR (LTHR)

91–100% **95–100%**

Just below the acid. Talking is hard, breathing is deep and rhythmic. One of the two key sessions of the week.

WHAT IT TRAINS

Raises FTP, improves lactate clearance and tolerance for sustained effort.

SESSIONS

1. **The 2 x 20** — 2 x 20 minutes at 95–100% of FTP with 5' recovery. The most tested threshold session in cycling.
2. **Criss-cross** — 3 x 15 minutes alternating 2 minutes in the lower part of Z3 and 1 minute in the upper part.
3. **Base threshold intervals** — 4 x 10 minutes at 95% of FTP with 3' recovery.

WHAT NOT TO DO

Going over your watts in the first three minutes. The lactate you produce there is lactate you won't be able to recycle, and the session turns into an anaerobic slog that trains none of what you came for.

~5–10%



Low threshold

The band right around FTP

% FTP % HR (LTHR)

100–102% **100–102%**

The exact band around FTP. It's the pace of a 40 km time trial and of the bike leg of an Olympic-distance triathlon.

WHAT IT TRAINS

Familiarity with race pace and tolerance for holding it without going over.

SESSIONS — ONLY IF YOU RACE AT THIS PACE

1. **Olympic-distance power** — 2 x 20 minutes at 100–102% of FTP with 5 to 10 minutes recovery.
2. **Time-trial blocks** — 3 x 12 minutes in Zone Y with 4' easy.

WHAT NOT TO DO

Living here. A couple of watts above FTP for 20 minutes is sustainable; five aren't, and you won't feel the difference until minute 15.

Residual outside race-specific blocks



High threshold

Intensive threshold. The other key session

% FTP	% HR (LTHR)
102–110%	102–105%

Intervals above FTP, longer than a VO2max rep and shorter than a tempo.

WHAT IT TRAINS

Raises the threshold from above: it improves your capacity to tolerate and recycle lactate at watts you couldn't hold for half an hour.

SESSIONS

1. **Long reps** — 5 x 6 minutes at 105–108% of FTP with 3' in Z1.
2. **Over-unders** — 4 x 9 minutes alternating 2 minutes at 105% and 1 minute at 95%.
3. **Medium reps** — 8 x 3 minutes in Z4 with 90 seconds easy.

WHAT NOT TO DO

Cheating with cadence. If you end up pushing the watts at 60 rpm you're doing weights on a bike instead of stressing the cardiovascular system. Hold 90–100 rpm.

~5–8%



VO2max

Short reps, low volume, high cost

% FTP	% HR (LTHR)
110–150%	105–120%

Short reps, low volume, high cost. Sharp pain and uncontrolled breathing.

WHAT IT TRAINS

Expands the aerobic ceiling, pedalling force and maximal fibre recruitment.

SESSIONS

1. **Classic VO2** — 5 x 3 minutes at 115% of FTP with 3 very easy minutes.
2. **30/30 micro-intervals** — 3 blocks of 10 x (30 seconds at 115–120% / 30 seconds easy), with 5' between blocks.
3. **Fives** — 4 x 5 minutes at 110–113% with 5' recovery.

Neuromuscular work. 8 x 20 seconds maximal, well above 150% of FTP, with full 3-minute-plus recovery. **Prescribed by effort and duration, not by zone** — it's measured afterwards, by looking at the power peaks.

WHAT NOT TO DO

Cutting the recoveries. If you reach rep 4 unable to repeat the watts from the first, you've stopped doing VO2max and started accumulating fatigue as a hobby.

~2–5%, including the neuromuscular work

Swimming

Primary metric: pace relative to your critical swim speed (CSS)

Water neutralises heart rate: it's imprecise and you can't look at it while swimming anyway. Everything is set by pace relative to your critical swim speed and the pace clock.

ZONE	% OF YOUR CSS	IF YOUR CSS IS 1:30/100M	WHAT IT'S FOR
1 · Recovery and technique	75–84%	1:47 – 2:00	Technique, easy swims, warm-up
2 · Aerobic base	84–91%	1:39 – 1:47	Where 80% of your volume lives
X · High aerobic	91–96%	1:34 – 1:39	Open water and 70.3 pace
3 · Tempo	96–100%	1:30 – 1:34	Extensive threshold. Key session
Y · Low threshold	100–102%	1:28 – 1:30	The band right around CSS
4 · High threshold	102–106%	1:25 – 1:28	Intensive threshold
5 · VO2max and speed	106–140%	1:04 – 1:25	Short reps, high cost

Why this is in percentages and not “CSS + 5 seconds”

Seconds per 100 don't transfer between swimmers. Zone 2 is **+7 to +14 seconds** over CSS if you swim 1:15, and **+12 to +23 seconds** if you swim 2:00. The same “+5 seconds” is Zone 2 for one and Zone 3 for the other. The calculator does that arithmetic with your number; the table is the part that doesn't change.

1	Recovery and technique	% OF YOUR CSS	IF YOUR CSS IS 1:30/100M
	Technique, easy swims, warm-up	75–84%	1:47 – 2:00

Very slow, very deliberate swimming. The day you're not there to train the engine, you're there to train the hand.

WHAT IT TRAINS

Feel for the water and technical correction.

SESSIONS

1. **Technique block** — 8 x 50 metres of drills (blind spot, single arm, closed fists) with a lot of rest.
2. **Warm-up and cool-down** — 200 to 400 easy metres at the start and end of every session.

WHAT NOT TO DO

Holding your breath or swimming tense. It should be a liquid massage.

~20–25%



Aerobic base

Where 80% of your volume lives

% OF YOUR CSS

84–91%

IF YOUR CSS IS 1:30/100M

1:39 – 1:47

Continuous, fluid pace, clearly slower than your CSS. **This is the zone most people swim too fast**, because in a pool the comfortable pace feels a lot like threshold pace.

WHAT IT TRAINS

Upper-body muscular endurance, mechanical efficiency, aerobic base.

SESSIONS

1. **Long broken swim** — 3 x 400 metres in Z2 with 20 to 30 seconds rest.
2. **Endurance pull** — 1000 continuous metres with a pull buoy, to isolate the stroke.
3. **Aerobic ladder** — 400 / 300 / 200 / 100 in Z2 with 20 seconds, repeated twice.

WHAT NOT TO DO

Racing the person in the next lane. It's the number-one error in pools and it converts all your aerobic volume into Zone 3.

~55–60%



High aerobic

Open water and 70.3 pace

% OF YOUR CSS

91–96%

IF YOUR CSS IS 1:30/100M

1:34 – 1:39

Open-water pace and the pace of a 70.3 swim. A little below CSS, sustainable for a long time, and with the same problem as in the other sports: it accumulates fatigue without giving what threshold gives.

WHAT IT TRAINS

Specific endurance at long-course race pace and holding a constant rhythm without wall references.

SESSIONS — OPEN WATER / 70.3 SPECIFIC

1. **Long blocks** — 4 x 300 metres in ZX with 20 seconds rest.
2. **Segment simulation** — 2 x 600 continuous metres in ZX with 45 seconds.

WHAT NOT TO DO

Using this zone as though it were Z2 because it feels comfortable. It gets prescribed on purpose or it doesn't get swum.

0% outside an open-water or 70.3 block



Tempo

Extensive threshold. Key session

% OF YOUR CSS

96–100%

IF YOUR CSS IS 1:30/100M

1:30 – 1:34

Your threshold pace. It demands real concentration: you have to become a metronome.

WHAT IT TRAINS

Raises the lactate threshold in shoulders and lats, and beds in race pace.

SESSIONS

1. **The CSS set** — 10 x 100 metres pinned to your CSS pace with 10 to 15 seconds rest. The short rest is the stimulus.
2. **Pace swim** — 4 x 200 metres in Z3 with 20 seconds.
3. **Long threshold blocks** — 3 x 300 metres in Z3 with 30 seconds.

WHAT NOT TO DO

Stretching the rest at the wall. The stimulus depends on it staying short: rest a minute between 100s and the set stops being a threshold set.

~5–10%



Low threshold

The band right around CSS

% OF YOUR CSS

100–102%

IF YOUR CSS IS 1:30/100M

1:28 – 1:30

The razor-thin band just above CSS. In swimming it's the pace you close a 1500 with.

WHAT IT TRAINS

Familiarity with the closing pace of a race.

SESSIONS — ONLY IF YOU RACE AT THIS PACE

1. **Closing a 1500** — 6 x 150 metres in Zone Y with 30 seconds rest.
2. **Rehearsal** — 3 x 400 metres: the first 300 in Z3 and the last 100 in Zone Y, without changing your technique.

WHAT NOT TO DO

Confusing it with Z4. It's two seconds per 100 apart and it feels very similar for the first 200 metres; the difference shows up at metre 600.

Residual



High threshold

Intensive threshold

% OF YOUR CSS

102–106%

IF YOUR CSS IS 1:30/100M

1:25 – 1:28

Clearly above CSS. The water gets heavy and your triceps burn.

WHAT IT TRAINS

Stroke power and tolerance for swimming fast with your technique still intact.

SESSIONS

1. **Short reps** — 15 x 50 metres in Z4 with 20 to 30 seconds.
2. **Reverse pyramid** — 200 / 150 / 100 / 50, accelerating as it shortens, with 30 seconds between reps.
3. **Hundreds** — 8 x 100 metres in Z4 with 30 seconds.

WHAT NOT TO DO

Letting the technique fall apart. In water, more brute force without technique is more drag, not more speed. If you feel like you're slapping the water, drop to Z3.

~5–8%



VO2max and speed

Short reps, high cost

% OF YOUR CSS

106–140%

IF YOUR CSS IS 1:30/100M

1:04 – 1:25

Your maximum pace over 100 and 200 metres, and above that, pure speed.

WHAT IT TRAINS

Maximal neuromuscular recruitment and an explosive catch.

SESSIONS

1. **VO2 set** — 8 x 100 metres hard with 45 seconds to 1 minute rest.
2. **Sprints** — 8 x 25 metres flat out, leaving every 1:30.
3. **Dive sprints** — 6 x 50 maximal metres with 2 minutes full rest.

WHAT NOT TO DO

Swimming the sprints tired at the end of the session. Speed is trained rested; otherwise you're training something else under the wrong name.

~2–5%

BONUS

The transition (the brick)

In triathlon the zones are affected by accumulated fatigue, and the metric rule changes there. Running in “Zone 2” straight off the bike ignores that the physiological cost of the bike has already lifted your heart rate: the pace that felt easy yesterday is metabolically expensive today.

In a brick, **heart rate outranks pace** — it's the only one of the two that knows what happened in the previous three hours. If your pulse climbs to Z3, slow the pace down even if the watch says you're going easy. It's the one situation in this whole guide where heart rate wins.

Who we are

Triaperformance is **Iván Koch's** coaching practice: triathlon and running training built on data, not on templates. One service, the same methodology for everyone — the one you have just read — and a weekly review of what you did and what's coming.

We work remotely with athletes across the Americas and Europe, from a first 10k to Ironman.

45

REVIEWS · ALL 5 STARS

328

PUBLISHED PLANS

3

LANGUAGES

“He provides insight into my Training Peaks data helping me understand what the numbers mean. He keeps my FTP current.”

Sylmarie Arizmendi · Puerto Rico

“Always on top of my training and available to make any adjustments I needed due to my travel schedule.”

Giles Carmichael · United Kingdom

THE NEXT STEP

You know which session grows each zone. Now they have to be ordered into a week.

These sessions work when they're sequenced: two quality sessions a week, everything else easy, and a progression that changes depending which month of your build you're in. Three ways to make that happen.

RECOMMENDED

US\$ 39.99/month

All-Access Membership

Every training plan in the catalogue, with these zones and these sessions already sequenced week by week. Switch plans as often as you like.

- All 328 plans: running, cycling, triathlon, HYROX
- TrainingPeaks Premium included
- Every guide and the full tools library
- No lock-in — cancel whenever you want

Start with All-Access

IF YOU'D RATHER WE DECIDED

US\$ 149/month

1:1 Coaching

A plan written for you and reviewed every week. Adjustments when life gets in the way, analysis of your sessions, and WhatsApp access. It's the service those 45 reviews describe.

See how it works

ONE SPECIFIC RACE

from US\$ 19.99

Single training plan

One goal, one plan, one payment. Choose by distance, weeks and level.

Browse the plans

Not sure which one fits? Email us at coach@triaperformance.com and we'll tell you honestly which one applies — including if it's none of them.